

The Ephemeris



March 2024

Volume 35 Number 01- The Official Publication of the San Jose Astronomical Association

March 2024 - June 2024 Events

Schedule subject to change

Please check our website for up to date information.

Board & General Meetings

Saturday 3/23, 4/20, 5/18, 6/22

Board Meetings: 6 - 7:30pm

General Meetings: 8:00 - 9:30pm

Solar Sunday (2:00 - 4:00pm)

Sunday 3/3, 3/31, 5/5, 6/2

In-Town Star Party (7:00-9:00pm)

Friday 3/8, 3/29, 5/3, 6/7

RCDO Starry Nights Star Party (7:00 - 9:00pm)

Saturday 3/2, 4/6, 5/4, 6/1

Imaging SIG (online)

Tuesday 3/19, 4/16, 5/21, 6/18

Binocular Star Gazing (at Coyote Valley OSP)

Saturday 6/1, 6/29

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Bogeyman

Photo by SJAA Member: Francesco Meschia
Nov. 16 - 18, 2022 Feb. 10 - 12, 2024
Location: Mountain View, CA

Francesco Meschia
ASTROPHOTOGRAPHY

SJAA Potluck and Awards Ceremony

We hosted our first potluck and in-person awards ceremony since the pandemic on February 24th, 2024! Over 20 attendees shared delicious drinks and food following the monthly board meeting. Dishes included meatballs, deli sandwiches, hummus, salad, and of course cookies and brownies for dessert, among others. Most importantly, several members were recognized during the awards ceremony for their outstanding contributions and volunteer work.

DOCTOR A.B. GREGORY AWARD

The Gregory Award is given in recognition of outstanding contributions of time and effort to others in amateur astronomy. Dr. Boris Gregory was a professor of French Literature at San Jose State College. He had a lifetime interest in astronomy, and for retirement, acquired one of the earliest C-8 telescopes. It was delivered personally by the then-president of Celestron, as he was a long-time friend of Dr. Gregory. After retirement, he taught an introductory astronomy class, part of the Adult Education series. It was in one of these sessions that Bob Fingerhut and other SJAA members met him.

Dr. Gregory was president of SJAA for one term, 1973-1974. He was on the Board from antiquity until his death in March 1979, and was especially adept at welcoming newcomers. With years of experience, he was often asked for advice and help, and his answer was typically, "Come over and we'll work on it". At board meetings, he was very often the first to say, "I'll look into that".

After his death, SJAA members Bob Fingerhut and Norm Neinchel conceived the Gregory Award in his honor. The award is normally given to only one person at a time, but this year saw two awards given to two extremely deserving members. Congratulations to **Rob Jaworski** and **Swami Nigam**!



Aroon presents Rob Jaworski with the Gregory Award. Rob currently serves as the Treasurer for SJAA. Photo credit: Ken Miura.

VOLUNTEER AWARD WINNERS

Renuka Veeraswami helps with our PRNS/ City of San Jose compliance program. She also has been helping with Solar Sundays and was a support person for the Annular Eclipse event.

Jonathan Lawton revamped the Quick STARt program and helps with the Loaner Program. He also volunteers at Starry Nights.

Hy Murveit runs the Imaging Special Interest Group and finds speakers for monthly online meetings/lectures/ demonstrations. The SIG is very popular and has a listing of 298 members.

Jordan Makowar is our astronomy educator extraordinaire! He regularly contributes articles to the Ephemeris, and he provides classroom education at elementary schools, both during School Star Parties and also during the day.

Richie Senegal had been a great help to firm up the policy documents for SJAA. His work has helped SJAA keep its charter charter and pass audits from the department of parks and rec.

Joe Fragola Has been leading the monthly Starry Nights program at RCDO, and he is a regular at ITSPs. He has also organized many member-only RCDO observing events.

Gary Mitchell has been a regular in our public star parties, as well as leader for School Star Parties for many years.

Mike Madden has contributed immensely in our public outreach events like ITSP, Starry nights, and Pinnacles.

Jon Anderson brings his telescope to schools all over the Bay Area. He is instrumental in the success of the SSP program.



Aroon presents Swami Nigam with the Gregory Award. Swami currently serves as the club's secretary. Photo credit: Emi Nikolov.



Dinner is served! Many members contributed to make the potluck a delicious success. Photo: Ken Miura.



Rob and Swami: 2024 Gregory Award winners! Photo: Ken Miura.



Aroon presents the volunteer award to Gary Mitchell. Gary's solar system models is a hit at star parties. Photo: Ken Miura.



Aroon presents Joe Fragola with the volunteer award. Joe works with OSA to put on star parties at RCDO. Photo: Emi Nikolov.



Joe, Emi, and friends enjoy good conversation and good food. Photo: Ken Miura.



Board members at the meeting prior to the potluck. The board meets every month to keep the club running smoothly! Photo: Emi Nikolov

Introducing Director 9

Credit: Marianne Damon

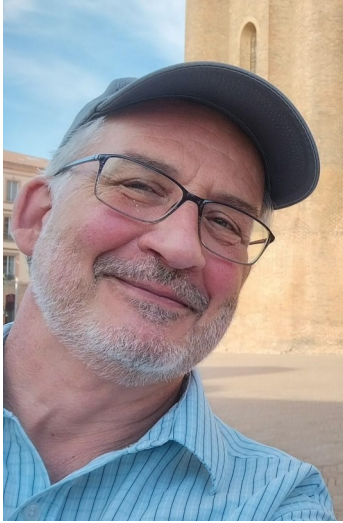


I am honored and pleased to accept the appointment to the Director 9 position on our Board of Directors. The position was vacated by Sukhada Palav who is now happily busy with New Mommie Duties. I will fill in for the remainder of her term. Many of you who attend our meetings at Houge Park know me as the person who brings the refreshments; my SJAA email name is “snacks”! Some of you know me from making astronomy-themed masks during the Pandemic. I have been a member since 2013; I signed up at my very first contact, a speaker night. I waited several years before buying any of my own gear, in the meanwhile taking advantage of Quick STArT and viewing through everyone else’s equipment until I decided on what would work for me. Guess what? I’m a bino gal, as I have trouble viewing with just one eye. Once I was set, I have helped with almost all our outreach events, including In Town Star Parties, School Star Parties, Starry Nights out at RCDO, Girl Scout events, Happy Hollow Star Parties, Cambrian Fest, Solar Sundays, Pinnacles Star Parties, and special events like our transit viewings. I also like to write and have contributed a few articles to the Ephemeris. I enjoy working with children and one of my first projects for our club was to make an astronomy-themed Activity Box. I’ve just recently completed a Planet Walk activity to use at schools as a filler in case clouds roll in. Including refreshments, my current “duties” include being a member of the 24” Dob Task Force (to find a home for our gorgeous big dob) and I’m also on the Compliance Team (maintaining City of San Jose requirements in return for use of Houge Park).

At school star parties I’m often asked how I became interested in astronomy. I answer that I cannot recall a time when I wasn’t. One of my earliest memories is sleeping on the porch in the Sierras in August, counting the shooting stars to over one hundred before falling asleep. My mother taught me the basic constellations. Sputnik 1 launched three days before my sixth birthday, and I so clearly remember the morning I saw Yuri Gagarin’s photo on the front page of the Mercury News, and was thrilled and inspired by the flight of “Seagull” Valentina Tereshkova. My early education flourished with a healthy balance of the arts and sciences (nowadays we have to re-invent it as STEAM). However, I was not able to really delve into astronomy until I retired from my career in health care as a physical therapist in clinical practice and management. In retirement I started birding and became a Master Gardener. I also am historian and associate editor for our local chapter of the Retired Public Employees Association (RPEA). But excuse me now, I have to start packing for my road trip to Texas for the Total Solar Eclipse. See you at Houge Park, or on the road to the next big astronomy adventure!

Introducing Speaker and Fix It Program Leaders

Introducing Larry Zurbrick— Fix It Program Leader



2024 marks the return of the SJAA Telescope Fix it program after a 2+ year hiatus. Larry Zurbrick has recently rejoined the SJAA after a few decade absence and has volunteered to take over the Fix It program from Vini Carter. Many thanks to Vini for his past year's coordination of the Fix It program!

Larry's background as an amateur astronomer includes 30 years of helping run public star parties at Santa Clara County's Grant Ranch Park and 25 years of being a Lick Observatory Summer program volunteer. He has a background in optics / semiconductor lithography (imaging) and is also an avid home shop machinist.

The Fix It program is targeted to serve **members** of the SJAA community who need help with collimation tasks and mechanical adjustments and minor repairs. The session is held every first Sunday of the month, from 2 to 4 PM (coinciding with the Solar Observing sessions) at Houge Park.

- **Appointments have priority for service.** Scheduling an appointment helps us to prepare what we need to bring in order to provide you with the best service possible, appointment sign up cutoff time is Sat noon before Sunday Fix It session

- **Fixing telescopes issues is the priority for Fix It sessions.** Training on how to use equipment *may be done if the appointment is scheduled and Fixit staff have adequate coverage.* The Fix It session is **not** intended to be a getting started in astronomy class.

- Sorry but, we are unable to disassemble telescopes to fix electrical issues with telescopes or disassemble telescopes to clean mirrors or corrector plates on the telescopes. (We may be able to give you instructions on how to do it so you can do it on your own.)

Please fill out the form on the SJAA website to schedule an appointment to help us be better prepared to assist you.

Introducing Leo Lipkin — Speaker Coordinator



I'm Leo Lipkin, a Firmware Test Engineer in the Bay Area with a background in Computer Science from UC Santa Cruz. My work spans embedded systems, hardware and software in fiber optics and wireless systems. Outside of tech, I enjoy outdoor activities like hiking and biking. Astronomy has always been a passion of mine—I own an 8" Schmidt-Cassegrain and a Seestar S50. I'm also your new event/speaker coordinator, excited to connect us with great speakers. Additionally, I am eager to learn more about astrophotography and enjoy flying drones, which allows me to combine my love for technology and the great outdoors.

Radio Signals From Space

Credit: Jordan Makower

Sunspot numbers are high right now, so I thought there would be a better chance at hearing/contacting a distant (or any) station by tuning around the 10- and 15-meter bands. I wasn't disappointed. With even my relatively low power ICOM-703 slowly moving up and down those bands, I heard the characteristic squeal and then deeper, truer, voices that I sought. But I heard other sounds that were unidentifiable. Not the rhythmic noises from engines or usual static from nearby transformers, yet something!

Since one of my hobbies is astronomy, I began to do research, and entered a 'rabbit hole' of fantastic information. Stars are mostly composed of hydrogen plasma, which is heated by compression and convectively rises producing a dynamic magnetic field. Electrons accelerated through this field emit radio waves. Our Sun produces radio frequencies between 24.6 MHz to 25.8 GHz. Even our Earth produces radio frequencies, but much lower, down to a few hertz. Jupiter also produces radio emissions that are enhanced or focused by Io, one of its moons. This effect is discussed in detail in <https://radiojove.gsfc.nasa.gov>. The frequencies involved cover 10 – 40 MHz, and could be heard by amateur radio receivers, *in theory*. The reason why those last two words are added is that those receivers are built with Automatic Gain Control, which suppresses noise, and prevents us from hearing broadband radio signals from stars and planets (like Jupiter).

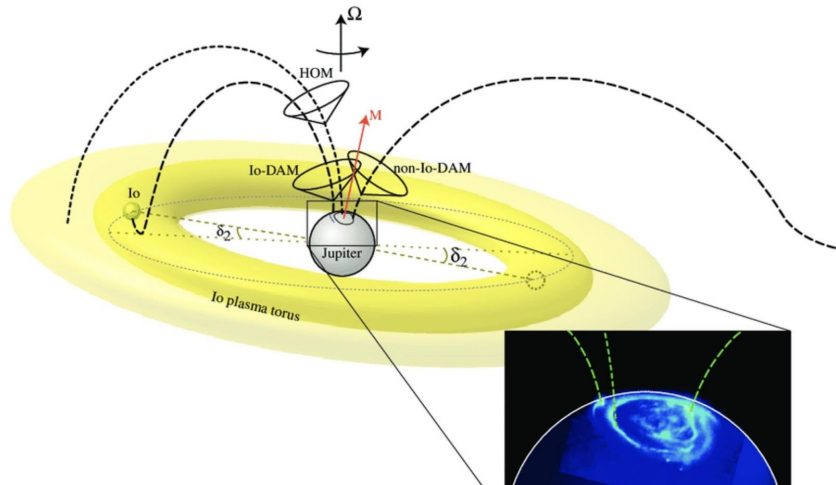


Diagram from Natural radio emissions of Jupiter as interferences for radar investigations of the icy satellites of Jupiter, Planetary and Space Science January 2010, DOI: 10.1016/j.pss.2011.06.012)

There are ways around this problem. One is to buy a kit that includes an SDR radio, antenna, and software for about \$220 + shipping, from radiojove.net (or professionally assembled for about \$384 + shipping). Another way is to use your computer to see and hear the “Official Radio JOVE LIVE HF Radio Astronomy Spectrograph hosted by K4LED, at his Larry Dodd’s station, in Georgia ([YouTube.com/channel/UCTawz3MnMBwjz9ShhSCoygQ](https://www.youtube.com/channel/UCTawz3MnMBwjz9ShhSCoygQ)). The radio noise sounds like air escaping from a compressed gas cylinder. Yet another way is to search for “The Radio Jove Project NASA” and navigate to the ‘Observer’s Corner’ where you will see a display of a ‘radio event’ that took place on June 21, 2023. If you select ‘Live Feed’ you would see that Larry’s equipment is picking up at that moment.

The display will show (and give data for) the positions of Jupiter and the Sun and will indicate the ‘hot spots’ of radio noise emanating from Jupiter and Io.

Use your computer to view star charts showing Jupiter’s position in the night sky (whatsouttonight.com), Star Walk 2 (Application developer Vito Technology Inc.), or skyandtelescope.org. Astronomy, like Amateur Radio, is a huge topic. A person who studies either or both will never be bored. Good Luck.

SJAA Member Astrophoto Gallery



Image: IC 342
By: Karen Bieber
Dates: Feb 10, 2024
Location: Cow Hill Observatory, Morgan Hill
Telescope: Askar 120APO
Camera: ZWO ASI533MC Pro
Mount: ZWO AM5
Software: Adobe Photoshop, Pleiades Astrophoto PixInsight

Astrobin Link: <https://www.astrobin.com/full/bzcx10/C/>

SJAA Member Astrophoto Gallery

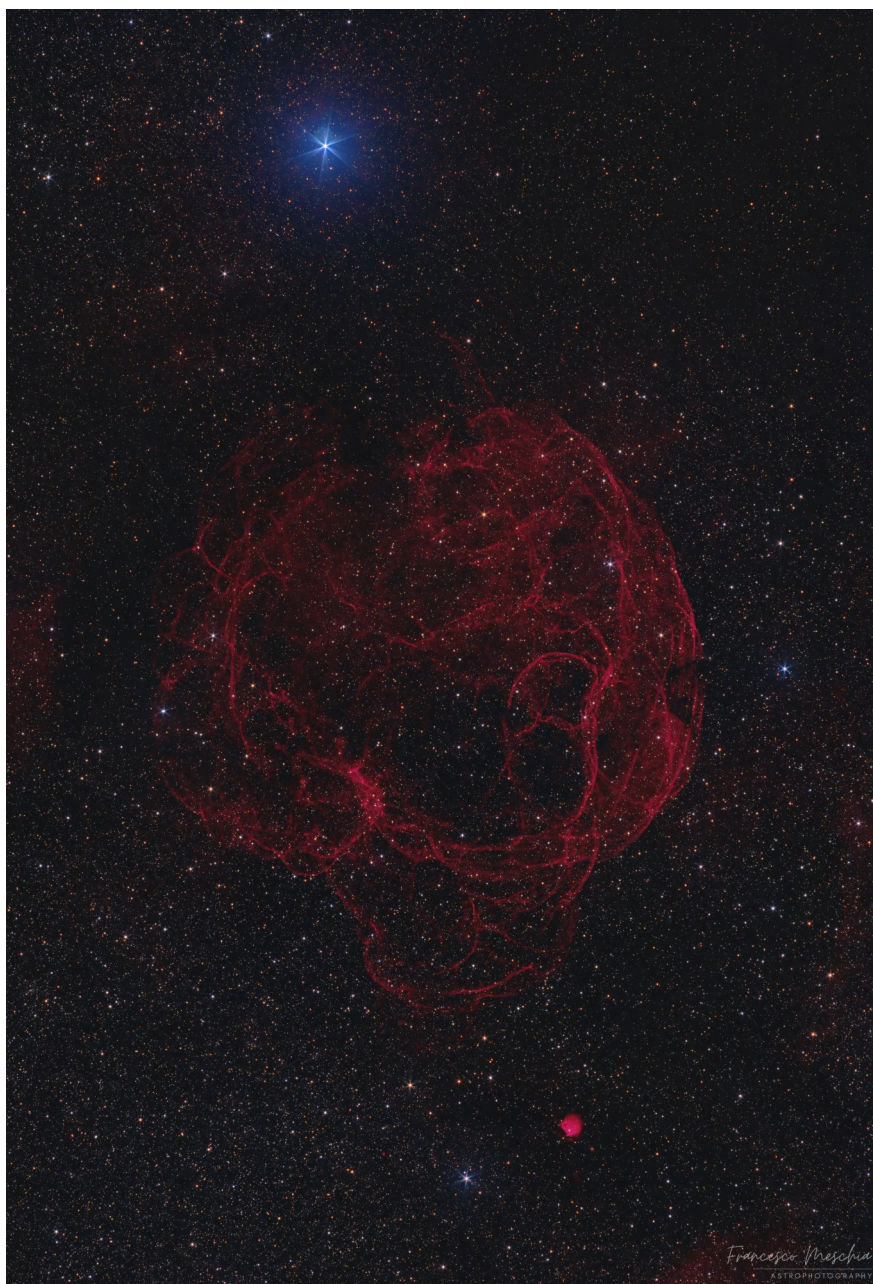


Image:	The Sharpless 240 / Simeis 147 supernova remnant
By:	Francesco Meschia
Dates:	1/3/2024, 1/5/2024, 1/6/2024, 1/11/2024, 1/15/2024
Location:	Home, Mountain View
Telescope:	Pentax SMC Super Takumar 135mm f/3.5
Camera:	ZWO ASI294MM Pro
Mount:	iOptron iEQ45 Pro
Software:	Adobe Photoshop, Cranfield / Payne Generalized Hyperbolic Stretching (GHS) Module/Script, Pleiades Astrophoto PixInsight, Russell Croman Astrophotography BlurXTerminator, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)
Astrobin Link:	https://www.astrobin.com/h7o6k1/

SJAA Member Astrophoto Gallery



Image: Rotten Fish Nebula
By: Swaroop Shere
Dates: 12/29/2023
Location: Stellar Skies Observatory, Pontotoc, Tx, United States
Telescope: Astro-Physics StarFire EDF "Gran Turismo" 130mm f/6.3
Camera: QHYCCD QHY268 M
Mount: Software Bisque Paramount MYT GEM
Software: Main Sequence Software Sequence Generator Pro (SGP) , Open PHD Guiding Project PHD2, Pleiades Astrophoto PixInsight, Software Bisque TheSkyX Imaging Bundle
Astrobin Link: <https://www.astrobin.com/full/apjddy/o/>

SJAA Member Astrophoto Gallery



Image: The Tadpoles Nebula
By: Hy Murveit
Dates: 12/27/2023
Location: Home, Silicon Valley
Telescope: GSO 10" f/8 Ritchey-Chretien Truss Tube
Camera: ZWO ASI1600MM Pro
Mount: Astro-Physics 1100GTO
Software: Adobe Lightroom Classic, KEDu KStars, Pleiades Astrophoto PixInsight, Russell Croman Astrophotography BlurXTerminator, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator

Astrobin link: <https://www.astrobin.com/ey2ocf/>

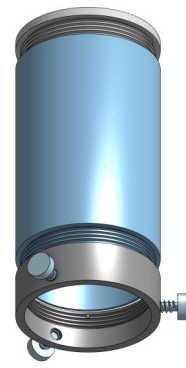
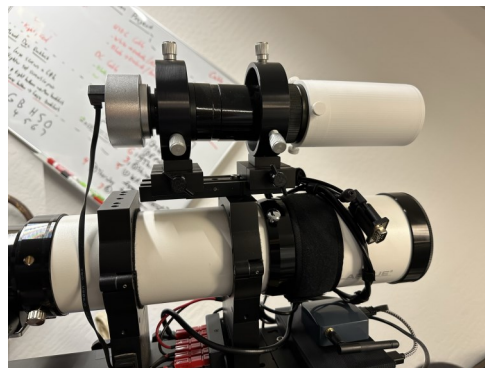
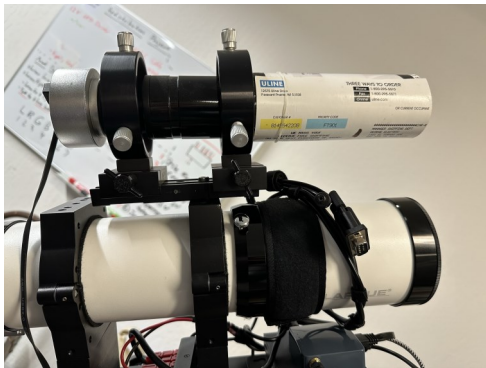
Fight the Dew

Credit: Carl Svensson

Dew might as well be a four-letter word for amateur astronomers. It sneaks up on you at the most inconvenient of times, and quickly ruins any observing session. These water droplets form on an object when that object's surface temperature falls below the dew point: the temperature at which the relative humidity is 100%, and water vapor in the air condenses into water droplets. Our telescope lenses and mirrors are particularly susceptible to dew, since they are pointing upwards at a cold, night sky that lowers their temperature faster than objects on the ground. Since dew is caused by an object cooling, a good way to prevent dew from forming is to raise the temperature of said object.

There are several methods to turn up the heat, and combat dew on your telescope lens. The most effective is to use a heating element, wrapped around the circumference of your scope near the lens. Astronomy shops call these "dew heaters." They come in a variety of sizes for a variety of scopes and require a 12v or 5v power source. These can be expensive, and of course require precious electricity, which can complicate a rig very quickly. They are often used with controllers that vary their output to conserve energy.

A simpler alternative is to use a dew shield. This is a tube that slides over the front of your telescope, and extends away from it for some length. The shield provides a bit of insulation to retain some of the scope's daytime heat and keep the dew at bay. It won't be 100% effective in all conditions, but it is often enough to prolong an observing session to a satisfactory length. And best of all: no power required. The shield also serves to block stray light from entering your telescope.



Left: Guide scope with paper-based dew shield. Middle: 3D-Printed dew shield. Note the dew heater on the primary telescope. Right: 3D Model of the printed dew shield.

You could buy a dew shield, but you can also make your own, and for small scopes, it is surprisingly easy. In my astrophotography rig, I use dew heaters on my primary telescope, but I also use a smaller, 50mm guide scope, for which I don't have a dew heater. I noticed that the dew would obscure the guide image enough to render it useless at a certain point in the night. Determined to make the most of every minute of clear sky, I set about to solve the problem with the latest technology: a cardboard tube. I tore off the back cover of a ULine catalog, cut it in half, rolled it up, fit it over my guide scope, and secured it with a rubber band. Voila. Instant dew shield. It worked like a charm, and got me through the tail end of Winter 2020.

Then, I got a 3D printer. After a few iterations, I landed on a neat design that doubles as an extendable dew shield and cap. This model provides a sturdy and modular dew shield for a 50mm guide scope. When not in use, the screw-on cap provides protection for the lens. I use it on my Orion 50mm Guide Scope with great success. The inner diameter of this model is 58mm, but the design could be adapted for larger apertures, as well. The shield is composed of three pieces: a base, middle, and a cap. These three pieces thread together, and the middle piece can be printed multiple times and screwed into other middle pieces to extend the dew shield. The dew shield fits over the business end of your scope, and is secured to the scope by three set screws, which are included in the model. Screw them in to tighten the fit against your scope. The walls of the shield are 4mm thick, which provide a sturdy and rigid body that won't flex or bend. The model prints without the need for supports, and should be printed in PETG or stronger material: PLA will melt and deform in a hot garage (trust me, I know!). The model is available for free on Printables, and I am happy to help anyone adapt it to their own scopes. Comment or send me a message on Printables: <https://www.printables.com/model/763134>

Dew can be a major pain, but with as little as a piece of heavy paper, it can be managed and even eliminated. It doesn't have to break the bank if you get creative. Good luck, and clear skies.

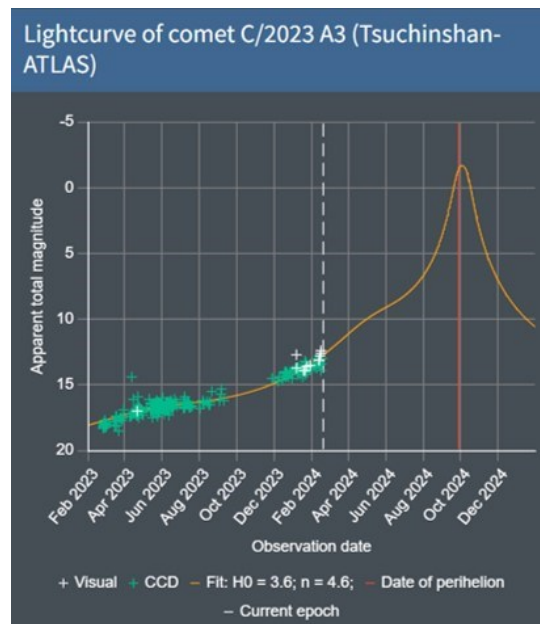
The comet C/2023 A3 (Tsuchinan-ATLAS) could be the next Hale-Bopp

Credit: Nikola Nikolov

C/2023 A3 (Tsuchinan-ATLAS) comet originates from the Oort cloud and is expected to become visible to the naked eye in October 2024. The comet will pass perihelion on September 27, 2024 at 0.39 AU from the sun. The closest approach to the Earth will be on October 12, 2024.



Currently the comet is in the constellation of Libra and it has an observed magnitude of 13, which makes the comet too faint for the naked eye and visible only by long exposure using Astroimaging rigs. Here are my first images of the comet made by a remote controlled telescope in Australia.



At the time the images were taken, the distance to the comet from the Earth was 3.33 AU, or about 450 millions kilometers (281 million miles), i.e. near Jupiter's orbit. The image above is a stack of three image frames, each with 5 minutes exposure time. As one can see, the comet already has a distinguishable tail. The predicted live light curve of the comet is shown below.

OBSERVING HIGHLIGHTS

3/20

Spring Equinox

At 03:01 UTC the Sun will shine directly on the equator: the Vernal Equinox.

3/25

Penumbral Lunar Eclipse

The Moon passes through the Earth's partial shadow, or penumbra.

3/24

Mercury Greatest Elongation

This is the best time to view Mercury, at an eastern elongation of 18.7 degrees.

4/8

Total Solar Eclipse

The last chance to see a total solar eclipse in North America until 2045.

4/22

Lyrid Meteor Shower Peaks

Dust from comet C/1861 G1 Thatcher is visible annually from the 16th to the 25th.

5/6

Eta Aquarid meteor shower

Comet Halley puts on a meteoric show April 19 - May 28, peaking on May 6th.

5/9

Mercury Greatest Elongation

At western elongation of 26.4 degrees, it's the perfect time to see Mercury.

6/20

Summer Solstice

At 20:46 UTC, the Sun will be directly over the Tropic of Cancer.

New Moon: 3/10, 4/8, 5/8, 6/6

Full Moon: 3/25, 4/23, 5/23, 6/22

Source: <http://www.seasky.org/astronomy/astronomy-calendar-2024.html>

Word Search

- ▶ Jupiter
- ▶ Pluto
- ▶ Moons
- ▶ Solar System
- ▶ Earth
- ▶ Satellite
- ▶ Titan
- ▶ Ganymede
- ▶ Mars

H A G P X U H Z I A O R M I D
P F L A L S K B Z X S M C W S
X X V A N Z X S T L U K Z B Y
S Q B O P Y P H J P O M O Y Y
Y S P B P D M W L T R T D N G
H A I X K V L E I P P Q D G T
Y D A D R P T E D K I N O K E
J M E Z F C D R H E C B W A J
M O N T O S A T E L L I T E U
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P G Q M N A R M J F R T Q N R
V V F V E A R T H Z A Y Q N B

Fun Facts

- ▶ Jupiter and Saturn compete for the most moons in our Solar System. Jupiter has 95 and Saturn has 146!
- ▶ New moons for both Jupiter and Saturn are still being discovered!
- ▶ Earth has only one moon, but it is over 1,000 miles wide!
- ▶ Jupiter's moon, Ganymede, is the largest in the Solar System. It's bigger than Mercury!

Source:
<https://science.nasa.gov/>

From the Board of Directors

Announcements

- ◇ Rob Jaworski, Glenn Newell and Wolf Witt were reelected into their respective Board of Director positions. Radhika Gupta was elected as a board member. Their term will expire in February 2026.
- ◇ New officer elections for President, Vice-President, Treasurer and Secretary positions will be held online via Electionbuddy. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.
- ◇ Current SJAA Members can benefit from signing up for the "SJ Astronomy" group on Meetup (<https://www.meetup.com/sj-astronomy/>) as well as the "SJAA Observers" Google group (<https://groups.google.com/g/sjaa-observers>).
- ◇ New Speaker Coordinator is Leo Lipkin
- ◇ Open positions: Librarian and Astronomy 101

Board Meeting Excerpts

January 27, 2024

In attendance

Aroon, Swami, Rob, Vini, Ken, Chris, Glenn, Peter,

Absent: Wolf

Guests: Emi, Marianne, Larry, Michael, Leo, Carl

Meetup plugin to automatically import events to SJAA website

Rob made a motion to approve spending no more than \$50 to purchase the plugin requested by our webmaster. Peter seconded. All approved, no abstentions or objections. Motion carried.

A/V setup for guest speaker presentations

SJAA task force lead by Arun will prepare a proposal to be presented to the board at the board meeting on Feb 24.

New Fence Banner for SJAA

Vini made a motion to approve spending no more than \$425 to get 2 new banners made. Marianne seconded. All approved, no abstentions or objections. Motion carried.

February 24, 2024

In attendance

Aroon, Ken, Wolf, Swami, Rob J, Peter, Marianne

Absent: Vini, Chris

Guests: Joe, Emi, Nancy, Carl S, Rich, Radhika, Larry, Carl W, Gary

Board member Elections

Results were announced for the Board of Directors Elections. 76 of 226 ballots were cast and there were 0 abstentions. There were 4 candidates for 4 vacancies - Wolf Witt,

Rob Jaworski, Glenn Newell and Radhika Gupta. All incumbents were reelected with Radhika Gupta as a new candidate. Their term is until February of 2026.

Officer Elections

SJAA Officer elections are to be held March 16-23. Swami to send out a general reminder to all members seeking positions.

Recruiting for Open Positions

Leader positions for 'Astronomy 101' and 'Library' are open, and SJAA is accepting leader applications.

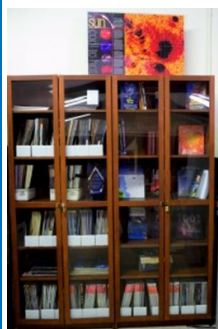
Simplified A/V setup for Hogue Park

Marianne made a motion to approve spending up to \$3000 (including taxes) to purchase the equipment requested above by the working group. Glenn seconded. All approved, no abstentions or objections. Motion carried.

Ephemeris Printing

Rob made a motion to change the front/back covers to color, and move the membership form to an inside page, without adding a card insert. SJAA will absorb the additional cost for now. Glenn seconded. Peter abstained, all others approved. Motion carried.

SJAA Library



SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program, is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Larry Zurbick, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held monthly. The next Quick Start session during this time period is March 5th at Houge Park and March 18 at RCDO.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by Neeti Song, each with a similar format, with only the content changing to reflect what's currently in the

night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group, this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

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SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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